

Evaluating Energy Consumption in CDN Servers

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Abstract. Energy management in large scale distributed systems has an important role to minimize the contribution of Information and Communication Technology (ICT) industry in global CO₂ footprint and to decrease the energy cost. Content Distribution Networks (CDNs) are one of the popular large scale distributed systems, in which the client requests are forwarded towards servers and are fulfilled either by surrogate servers or by the origin server, depending upon the contents availability and the CDN redirection policy. In this paper we explore the energy consumption in CDNs using different client redirection policies. We propose a new model to measure energy consumption in the CDN surrogates. The surrogate servers' utilization is used as a key criteria to measure the consumed energy. We show the impact of minimizing the number of surrogate servers on the energy consumption and on the other performance evaluation parameters in CDNs (e.g. servers utilization, mean response time, byte hit ratio). We also exhibit, how the energy consumption and the other performance evaluation parameters are affected by the change in the number of content requests, i.e. by the traffic load.

Keywords: Content Delivery Networks, Surrogate Servers, Energy Consumption.

1 Introduction and Motivation

ICT contributes more than 2% in the global CO₂ footprint that is increasing day by day [1]. Public and private organizations are interested to find the less energy consuming solutions in order to protect the environment and to minimize the overall cost [2]. Industrialists are introducing the new products in the market with the Green tag. According to [3], 15% of data centers' cost is in terms of its energy consumption. One of the popular large scale distributed systems Akamai, is estimated to spend a \$10M of electricity cost every year [4]. The Internet has made the communication faster and easy. The Internet has gotten mature to the point that web browsing and e-mail are no longer the main features. Content providers and e-commerce organizations view the Internet as a tool to make available their rich contents to their customers. Therefore, there is very fast increase in the Internet infrastructure. Internet providers are installing more energy consuming devices in order to provide better services. Energy cost

of the Internet infrastructure is increasing every year, that affects the both organizations and their customers as well. To explore the energy consumption and its reduction in large scale distributed systems is a hot research area [5] , [6] with important practical applications.

A popular type of such a network is the Content Distribution Network (CDN) [7], which is responsible for managing the large amount of content traffic originating from the Web users. A CDN consists of a set of surrogate servers geographically distributed in the Web, which contain copies (replicas) of content belonging to the origin server (according to a specific storage capacity). The main idea is to bring content replicas closer to the user. Therefore, CDNs act as a network layer between the origin server and the user, for handling their requests. With this approach, content is located nearer to the user, yielding low response time and high content availability since many replicas are distributed. The origin server is relieved from the requests since the majority of them are handled by the CDN servers. A typical CDN includes the following functionalities 1) Content distribution services, i.e., geographically distributed set of surrogate servers, which store the data on behalf of the origin server or using data replication. 2) Request redirection and content delivery services, e.g., the client requests are directed towards the closest servers either in terms of proximity or load. 3) Cooperation services, i.e., the surrogate servers may cooperate with each other in order to fulfill the client requests. If a client sends content request to a server, the server checks for the content in its cache: If the content is not available in its cache it forwards the request to the neighboring surrogates. 4) Management services, e.g., services to control utilization of the contents, managing copy rights etc. 5) Content adaptation services, e.g. format conversion.

Most of the cited works address how the underlying network activity in conjunction with content delivery interacts with the energy consumption, but there is not much focus on the CDNs domain in particular. Therefore, our key motivation lies on finding the energy consumption measures in the CDN servers. Note that, the energy cost of the network or the transport energy cost is not taken into account in this paper. We take the load of the CDN servers as a key parameter to evaluate their energy consumption.

The main advances of this research are the following:

- Deriving an energy consumption model from the surrogate servers' utilization in a CDN.
- Studying the impact of the number of surrogate servers, participating in the CDN, in terms of energy consumption and quality of services parameters.
- Studying the impact of the number of client requests in terms of energy consumption and quality of services parameters.
- Deriving from these studies some perspectives as potential energy saving techniques that preserve quality of services while respecting a certain level of quality of services for the client requests.

In the next section, CDN request redirection policies are presented which show how the client requests will be forwarded to the surrogates. In section 3 CDN utilization and energy consumption models are proposed. Section 4 describes

the simulation testbed. Section 5 states the performance evaluation where our proposed methods are evaluated and explained with the help of different graphs. In Section 6, we discuss some previous related works. Section 7 concludes the paper.

2 Towards Energy Saving Policies in CDN

CDNs occupy a considerable amount of Internet traffic. CDN plays an important role to distribute the real time media traffic to the widely dispersed Internet users. There is intensive need to introduce energy-aware techniques and technologies in the CDN infrastructure in order to save the environment and the money as well. Different kinds of traffic demand different level of services e.g. real time media traffic requires the minimum delay. The purpose is to develop methods and techniques to reduce the energy consumption in the CDN while not much compromising on Quality of experience. Various techniques are proposed for energy efficiency in networks such as described in Section 6. One of the popular approach is to redirect the traffic towards fewer devices and to shut down the others or to put them in sleeping mode. The concept is based on the fact that the network traffic is not always in the same manner. In normal network conditions, network devices are not utilized according to their full capacity. There is always a need to introduce the smart mechanisms which permit to utilize the network devices according to their capacity and to gain the energy savings in an efficient way.

We consider the scenario of a CDN and propose two policies which redirect the client requests to the CDN surrogates. We use the Zipfian distribution in order to define these policies. In the start, the surrogates are assigned requests randomly then the CDN surrogates are sorted according to their current utilization. Finally the surrogates are picked according to the probability drawn from the respective Zipfian distribution with the slope parameter z .

2.1 Random

For the first policy, that is called Random policy, the requests are sent to the surrogates randomly. For random policy Zipfian parameter z is set to 0 that creates the uniform distribution. According to this policy all the surrogate servers have the equal probability to serve the client requests. So, the content requests from the clients to the CDN servers can be redirected to any of them. Therefore all the surrogates have the equal chance to get the client demands throughout the execution time. This policy is better in the case when high availability of the servers is required.

2.2 Loadunbalance

For this policy we set the value of Zipfian parameter z to 1. In this policy the requests are distributed in an exponential fashion. According to this policy,

most of the client requests are redirected towards a certain number of surrogate servers while the other surrogates have the less priority to get the requests. A small number of the surrogates capture most of the client requests traffic. So, the most of the CDN servers are underutilized, which can be considered to use the techniques for the energy savings. A threshold is arbitrarily set to limit the maximum load of each surrogate server of its full capacity (this is set to 90% in our experiment).

Please note that in this paper, these policies are just considered to redirect the client requests. In other words, we do not want to investigate in this paper advanced techniques, such as, of switching off nodes when they have a low load (this is future work), but we rather study the impact of using less number of servers on quality of services parameters (as a function of the load) and also the expected savings in terms of energy (modeling the energy consumption as a function of the servers' utilization). We believe this step is necessary in order to motivate future work on energy savings.

3 Modeling CDN Servers' Utilization and Energy Consumption

When a surrogate server has no request to serve then it is considered as idle. When it gets a connection in the form of content request from a client or from another surrogate server (in case of cooperation), it is said to be utilized. A surrogate server can have multiple connections at the same time, depending upon its capacity. Its utilization is directly proportional to the number of connections it handles at a given time. If the ratio of the number of current connections to the maximum number of possible connections is larger, the surrogate server is said to be better utilized and vice versa. The connection duration is important to calculate the utilization of a surrogate server since some requests may take different durations to be served (depending typically on the load of the server but also of the size of the data being requested).

A connection to the surrogate server s is established 1) when a client c makes the request for contents to s ; or 2) when another surrogate server s_2 requests the contents from s in case of cooperation.

3.1 CDN Server Utilization Model

Here, we present a simple CDN server's utilization model. We first, compute the utilization ratio of the server s during the time interval $[t_1, t_2]$ as such:

$$UR_{s[t_1, t_2]} = \frac{Conn_{s[t_1, t_2]}}{ConnMax_s} \quad (1)$$

where $Conn_{s[t_1, t_2]}$ is the actual number of connections the surrogate server s handles between time t_1 to time t_2 (considered as constant between t_1 and t_2). $ConnMax_s$ represents the maximum number of connections allowed on the

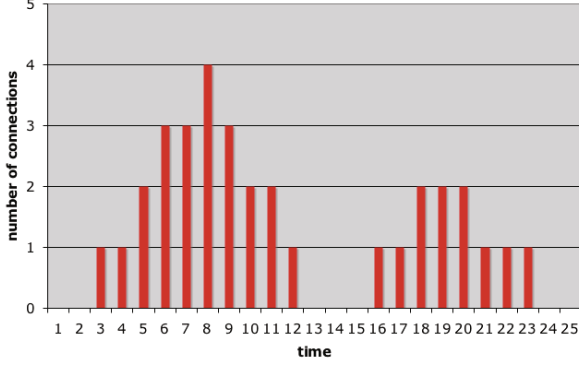


Fig. 1. Number of connections over time

server s i.e. the maximum content requests a surrogate server s can have at the same time. It shows the capacity of a CDN server.

During the lifetime of a server, its utilization ratio will increase and decrease over time, as shown in Figure 1. In this Figure, if we consider $ConnMax_s = 5$, we have $UR_{s[6,7]} = 3/5$. The duration of this utilization ratio is 2.

Hence, we can compute the utilization of a server s between t_i and t_j as:

$$U_{s[t_i, t_j]} = \frac{\sum_{k=i}^{j-1} UR_{s[t_k, t_{k+1}]}(t_{k+1} - t_k)}{t_j - t_i} \quad (2)$$

$$U_{s[t_i, t_j]} = \frac{1}{(t_j - t_i) * ConnMax_s} \left(\sum_{k=i}^{j-1} Conn_{s[t_k, t_{k+1}]} * (t_{k+1} - t_k) \right) \quad (3)$$

In the same example, we have thus $U_{s[0,12]} = (0 * 2 + 2 * 1 + 3 * 2 + 4 * 1 + 3 * 1 + 2 * 2 + 1 * 1) / (12 * 5) = 0,33$, meaning that during this period the server is used at 1/3 of its capacity in average.

Finally, the utilization of the server s during an experiment with a duration T is:

$$U_s = U_{s[0, T]} \quad (4)$$

From this utilization model one could derive an energy consumption model linking the utilization to the consumed energy.

3.2 Definition of Energy Consumption in CDN Servers

In CDNs the majority of energy consumption originates in surrogate servers. Each surrogate server consumes a constant quantity of energy just by being turned on. The rest can be considered proportional to the utilization. In this context, we assume energy consumption to be proportional to the ratio of active connections

against the maximum simultaneous connections each surrogate server is able to handle. These number of active connections accounts for the work being done at the server side to retrieve the data (handling of the index), the disk IO to fetch the data, the network connection and the cache management policies. An extended model could be used in order to derive power consumption based on actual load on IO, networks and CPU, and could be considered in future. However, even using a such basic assumption is sufficient, in order to compare the energy consumption of different configurations of number of servers and traffic requests.

3.3 CDN Server's Energy Consumption Model

On the basis of above described definition for energy consumption in CDN servers, we propose a simple model of energy consumption in surrogate servers. Surrogate server's utilization is used as a parameter in order to measure its energy consumption. First, we calculate the power consumed by the surrogate servers while serving the contents to the clients or to the neighboring surrogate servers (in case of cooperation). The power consumed by the surrogate server at a given time can be calculated as follows:

$$P_{s[t_1, t_2]} = P_{idle_s} + \frac{Conn_{s[t_1, t_2]}}{ConnMax_s} (P_{Max_s} - P_{idle_s}) \quad (5)$$

where P_s is the power consumed by the the surrogate server s . P_{idle_s} is the minimum possible power the surrogate s can consume. In this case when a surrogate server is turned on it is supposed to consume a constant amount of power if it is idle and doesn't have any request to serve i.e. it is completely unloaded.

Between time intervals t_i and t_j the energy consumption $E_{[t_i, t_j]}$ can be calculated as:

$$E_{s[t_i, t_j]} = \sum_{k=i}^{j-1} P_{s[t_k, t_{k+1}]} * (t_{k+1} - t_k) \quad (6)$$

$$E_{s[t_i, t_j]} = \sum_{k=i}^{j-1} (P_{idle_s} + \frac{Conn_{s[t_k, t_{k+1}]}}{ConnMax_s} (P_{Max_s} - P_{idle_s})) * (t_{k+1} - t_k) \quad (7)$$

$$E_{s[t_i, t_j]} = (t_j - t_i) * P_{idle_s} + \frac{1}{ConnMax_s} (P_{Max_s} - P_{idle_s}) \sum_{k=i}^{j-1} Conn_{s[t_k, t_{k+1}]} * (t_{k+1} - t_k) \quad (8)$$

Linking the energy consumption and the utilization model proposed earlier, we obtain:

$$E_{s[t_i, t_j]} = (t_j - t_i) * P_{idle_s} + (P_{Max_s} - P_{idle_s}) * U_{s[t_i, t_j]} \quad (9)$$

Finally the total energy E consumed by a surrogate server s is shown by the following equation:

$$E_s = E_{s[0,T]} \quad (10)$$

4 Simulation Testbed

In order to evaluate the energy consumption and quality of services of CDN over different configurations, we need a testbed that provides us the CDN analytical simulation environment because the CDN real time applications are hard to get for research purpose. This environment includes a) a system model simulating the CDN infrastructure, b) a network topology generator, c) a website generator, d) a client request stream generator.

4.1 System Model

A suitable simulation environment for this purpose is CDNSim [19]. CDNSim simulates a main CDN infrastructure and is implemented in the C++ programming language. It is based on the OMNeT++ library which provides a discrete event simulation environment. All CDN networking issues, like surrogate server selection, propagation, queuing, bottle-necks and processing delays are computed dynamically via CDNSim, which provides a detailed implementation of the TCP/IP protocol, implementing packet switching, packet retransmission upon misses, freshness, etc. CDNSim allows to add new client redirection policies. We consider the case of identical, geographically distributed surrogate servers where each server can handle 500 connections at the same time. The clients are divided into 100 groups distributed all over the world.

4.2 Dataset

We used a real Internet topology of AS-level, having 3037 routers, that consists of routing data collected from 7 BGP peers dispersed at different locations. The backbone network topology has a set of routers. The other network entities like surrogate servers, origin server and clients are connected randomly to the router backbone. A synthetic but realistic website having 50000 objects of 1GB total size, is generated. A request stream generator is used that takes the website graph and generates requests stream using random walks [20]. Not all the website objects are requested with the same frequency. Popularity of the objects in a website graph is considered using Zipfian distribution [20]. The higher values of the parameter z cause the handling of most of the requests to the smaller number of objects. As different objects in a website can have different popularity and size. There may and may not be a correlation between size and popularity. The correlation between size and popularity is considered to have the values in a range from -1 to 1. Negative values indicate that an object smaller in size will have more popularity than the larger ones and vice versa. The value 0 lies in between the two extremes where objects popularity is not related to the size of the objects.

5 Performance Evaluation

5.1 Evaluation Parameters

In order to proceed gradually, first we tested our approach with a warm-up phase of a 50000 requests of traffic (not shown here). After that we performed the experiments with the traffic of 1 million requests, which are evaluated here. The following measures have been taken into account.

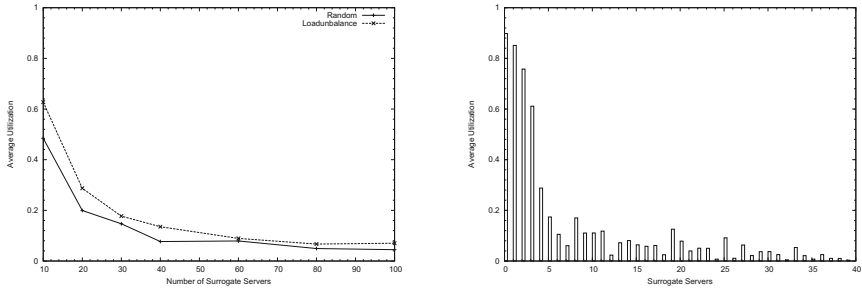
- *Servers' Utilization.* We compute the average of each surrogate servers' utilization. The values of surrogate servers' utilization range from 0 to 1 (see Section 3.1).
- *Energy Consumption.* It is the power consumed by a surrogate server or a set of surrogates during a time period. We evaluated energy consumption in joules (see Section 3.3).
- *Mean Response Time.* It exhibits the average user experience of the CDN. It is the ratio of the summation of the time taken to fulfill all client requests to the total request number.
- *Byte Hit Ratio.* It is used to measure network activity and bandwidth consumption. It is the hit ratio presented in bytes where hit ratio represents the ratio of the served requests to the total number of requests which are handled directly by a surrogate server without any cooperation with the other servers. We present it in percentage.
- *Failed Requests.* These are the requests made by clients for contents but the requests are not fulfilled even after ten retries. It is shown in percentage.

Experiments settings. In order to have a meaningful surrogate servers utilization, first of all we did an experiment with different number of surrogate servers, e.g. 10, 20.....100. For this purpose, we used two request redirection policies i.e. random and loadunbalance which have been already discussed in Section 2. Other simulation parameters are shown in Table 1 Experiment 1. We had two purposes for this experiment, 1) to choose a request redirection policy, 2) to make choice of maximum number of surrogate servers: for the further experiments.

The result of this experiment is shown in Figure 2-left, where the x-axis shows the number of surrogate servers while at y-axis the average utilization of the surrogates is presented under two request redirection policies i.e. random and loadunbalance. It is shown in Figure 2-left that, if we have the less number of surrogates, the average utilization is better than when we have a larger number of surrogate servers. This trend of utilization can be seen in case of both redirection policies where the decrease in average utilization curve is gradual with the increase in number of surrogates. But it can be noticed from the figure that the loadunbalance performs better than the random, in case of the average utilization of the surrogates. On the other side, we see that in case of loadunbalance, the average utilization of surrogates is less than 10% for 40 servers and in case of random, the average utilization is less than 10% after 30 servers. In the normal conditions in a real CDN network, the average utilization of the servers is between 10% to 15% to accept high burst loads. By analyzing these results, we

Table 1. Summary of simulations parameters

Parameter	Experiment 1	Experiment 2
Website size	1GB	1GB
Website number of objects	50000	50000
Website z for size	1	1
Size vs. popularity correlation	0	0
Number of requests	10^6	$10^5, 2 \times 10^5, 3 \times 10^5 \dots 10^6$
Mean interval time of requests	0.01sec	0.01sec
Distribution of the interval time	exponential	exponential
Requests stream z	1	1
Link speed	6Mbps	6Mbps
Network topology backbone type	AS	AS
Number of routers in network backbone	3037	3037
Number of surrogate servers	5,10,15...40,60,80,100	5,20,40
Number of client groups	100	100
Number of content providers	1	1
Cache size percentage of the website's size	20%	20%
Load-unbalancing paramter z value	0, 1	1

**Fig. 2.** Left: Average Utilization as a function of Number of Servers vs. Redirection Policies

Right: Average Utilization of 40 Surrogate Servers

make the choice of loadunbalance redirection policy for the further experiments and to try it up to 40 servers, where we have an utilization of 13.5%. Figure 2-right shows an example of the average utilization of the 40 surrogate servers with loadunbalance policy with 10^6 content requests.

We also computed the distribution of the 10^6 requests' durations when using 40 servers. More than 90% of the requests served are below the average response time. However, some requests (less than 0.1%) last much longer than others, increasing at the end the average request time.

5.2 Number of Servers vs. Evaluation Parameters

In this experiment we evaluate the impact of number of surrogate servers on the above discussed evaluation parameters in the normal CDN conditions where

the surrogate servers are not utilized according to their capacity. The simulation data set can be seen in the Table 1 in Experiment 1. We use the loadunbalance policy with the number of surrogates 5, 10, 15..., 40. We kept the other simulation parameters same as used in the earlier discussed experiment.

In figure 3-left, x-axis shows the number of surrogate servers from 5 to 40 while y-axis presents the energy in Joules consumed by the surrogates. We can see the gradual increase in the total energy consumption by these sets of surrogates with the increase in the number of surrogate servers and vice versa. The purpose behind this experiment is that if we apply the traditional technique of minimizing (by diverting the load to the fewer servers and switching-off the others) the number of surrogate servers in CDN normal traffic conditions, then what kind of impact we can have on the energy consumption, within CDN servers and on the other evaluation parameters. It should be noticed that, here we didn't take into account the energy consumption in the network. In the case of decreasing the number of servers, energy cost of transport can be increased which can increase the overall CDN energy cost. Figure 3-left shows that we have a considerable amount of energy savings by minimizing the number of surrogate servers. For example, if we use the 20 servers instead of 40 servers, the total energy consumption within the surrogates is decreased by 37%.

The obvious drawback is a loss in quality of services. With the decrease in the number of servers from 40 to 20, we witness an increase of 14% in mean response time as shown in Figure 3-right. That is important to notice for the user experience. It can be experienced due to the increase in distance between clients and servers. Congestion at the nodes near to the servers can also be resulted into higher values of mean response time. Figure 3-right also exhibits an inverse behavior between the byte hit ratio and the mean response time. With the increase in the number of servers, clients have the more number of the available surrogates that increases the byte hit ratio.

With the decrease in the number of surrogate servers, the load on the surrogates is increased, which impacts the failed requests as well. The number of failed requests is increased with the decrease in the number of surrogate servers.

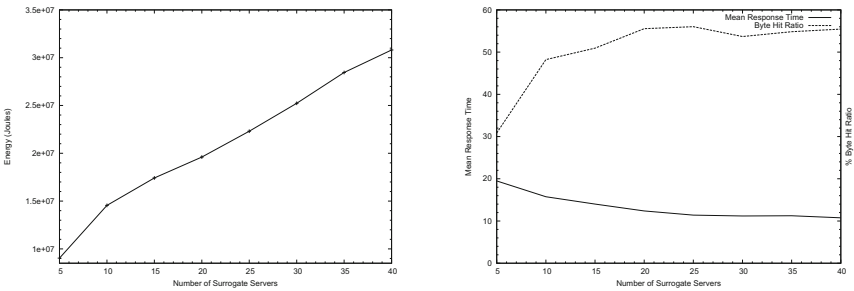


Fig. 3. Left: Energy Consumption vs Number of Surrogates
Right: Mean Response Time/Byte Hit Ratio vs Number of Surrogates

For example the failed requests in case of 5 servers are more than 8%, while with 20 surrogates, it decreases down to 0.78%.

Finally, while minimizing the number of surrogates, we are decreasing the CDN servers energy consumption but we have the cost of increased mean response time, decreased byte hit ratio and increased failed requests.

5.3 Number of Requests vs. Evaluation Parameters

In this experiment we analyzed how the number of client requests affects the evaluation parameters. The parameters of the simulation are shown in the Table 1 Experiment 2. In this experiment the number of requests used are between 10^5 and 10^6 . We use 5, 20 and 40 servers for this experiment. The loadunbalance policy is used to redirect the content requests.

In Figure 4-left the average utilization of the surrogate servers increases slowly (logarithmically) with the increase in the number of client requests.

The same kind of behavior can be seen in Figure 4-right where the energy consumed by the surrogates is increasing linearly with the increase in the number of requests. Indeed, the load of the servers is increasing with the number of requests to serve.

Figure 5 shows the impact of number of requests on the mean response time and on the byte hit ratio. It represents that the mean response time decreases gradually with the gradual increase in the number of requests and at the same time the byte hit ratio show the opposite behavior than the mean response time. For this behavior of mean response time and byte hit ratio, surrogate's cache plays an important role. When the clients make requests for contents, surrogate checks these contents in its cache. If the contents are available, it serves it to the client. In the case of unavailability of the contents, surrogate makes request to the other surrogates or to the origin server (depending upon the request redirection policy). Over time surrogate servers' cache becomes more intelligent and it tries to keep the frequently demanded contents in its cache. That's why with the increase in amount of client requests the byte hit ratio increases, which indicates the increase in the number of the directly served contents by the surrogate. That

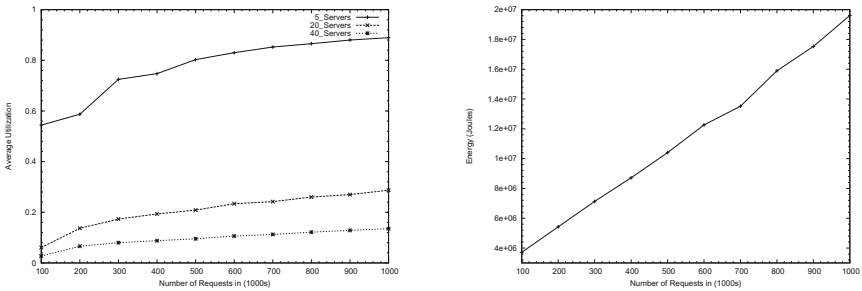


Fig. 4. Left: Surrogates' Average Utilization vs Number of Requests

Right: Total Energy Consumption in Surrogates (20 servers) vs Number of Requests

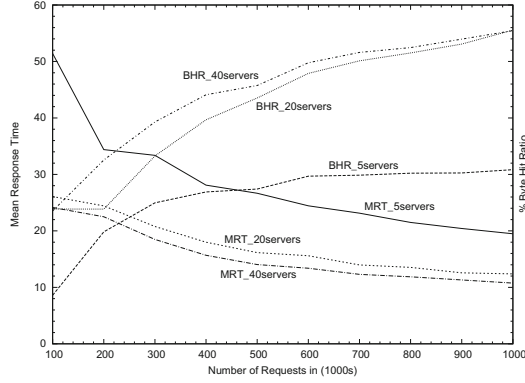


Fig. 5. Mean Response Time/Byte Hit Ratio vs Number of Requests for 5, 20 and 40 servers

decreases the cooperation among the surrogates and indirectly mean response time for the client requests decreases. In this experiment, the failed requests for 5 servers are noticed from 8% to 10% while in case of 20 servers, it is from 0 to 3.9% and in case of 40 servers, it is always lower than 1%.

We can conclude that increase in the number of requests increases the energy cost but at the other hand we experience the lower mean response time, higher byte hit ratio and lesser failed requests.

6 Related Work

CDNs have attracted the researchers to propose and implement new techniques to improve Internet infrastructure. Typically, a client request is forwarded towards the nearest surrogate server in CDN. There are some CDN-dependent decisions such as organization of the CDN, from where to pull the content on cache miss of a surrogate (e.g. policy decides either from other surrogate or from origin server), degree of cooperation among surrogate servers etc. So, it is important to define some essential data management issues. Till now, different policies have been defined in order to distribute the contents in CDNs. In [7], cooperative and uncooperative push-based policies are presented. [11], [12], [13] describe cooperative and uncooperative pull-based policies. In cooperative policies surrogate servers cooperate with each other on the cache miss while in uncooperative policies surrogate servers don't cooperate with each other. In push-based policy, the content is pushed from the origin server to the surrogate servers. In pull-based policy, clients' requests are directed (through DNS redirection) to their closest (in terms of geographic proximity or load, etc.) surrogate server.

In last few years energy conservation and energy efficiency in computer networks have gained a valuable attention. Different solutions have been proposed to reduce the energy consumption such as: Manufacturing hardware that follows some energy-efficiency standards [14]. [15] proposed to change the link rate

dynamically according to the traffic load, to gain energy savings. Diverting the traffic load to fewer links and make the other links idle to put them in sleeping mode [16]. [17] describes frequency change and Dynamic Voltage Scaling (DVS) for energy reduction of integrated circuits. All the above mentioned works are related to our work in the context of energy in networks but they differ in the context that our focus is on CDNs. A.Feldmann et al. [8] analyze the energy trade-offs among content delivery architectures. They took three content distribution architectures (data centers, P2P and CDN). They proposed an energy consumption model in the context of IPTV. They conclude that CDNs are clear winner in terms of total energy costs. In [18] a utilization-aware redirection policy for energy conservation in CDN is presented. It is based on a CDN utilization model but no model for energy consumption in CDN servers is proposed. [9] presents a constant power model. It doesn't take into account the resource utilization and predicts the constant power. This model provides the base for the utilization-based models. Load models concentrate on the running hosts only and are sufficient to predict the energy consumption for the servers which are active 24/7[10]. Load models take into account the utilization of the servers. After turning the servers on, their active energy consumption is the function of their load. We adopted this form of model which is fit in our case.

7 Conclusion and Future Work

CDNs have the capacity for energy reduction in order to protect environment and to reduce over all system cost. So, it is important to model the surrogate servers' utilization and its energy consumption. CDN servers' load can be used as a parameter to measure the energy consumption in surrogate servers. In this paper, we proposed a energy consumption model and we studied the impact of the number of servers and traffic on the energy consumption and quality of services parameters. The results show the actual potential for saving energy by using considerably less servers and tuning them to their actual load so that the quality of services are respected, and that some threshold have to be determined so as to guarantee these. Minimizing the number of surrogate servers can provide energy savings in normal CDN conditions but user performance parameters are also important to consider. In future, we plan to continue our research in this direction. Our next step is to divert the load to the fewer servers and to actually shut down the idle servers dynamically. A similar work is to change the processor frequency of the surrogate servers according to their load. Finally, we will have to consider advanced models for servers' energy consumption (and to see if these have an impact on the results) and we aim to analyze and model energy cost of network and data transfer in CDNs.

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